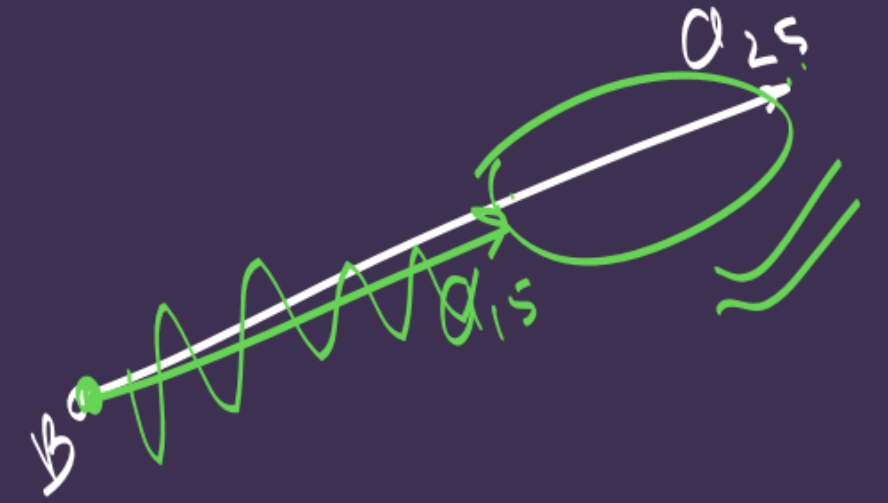


ii. Find the distance between the 15th pole and the 25th pole.

$$a_{25} - a_{15}$$

dist of 25th pole from base

dist of 15th pole from base



iii. a. Find the time taken by the cable car to reach the 15th pole from the top if it is moving at the speed of 5 m/sec and coming from the top. (2)

OR

b. Find the total number of poles installed along the entire journey. (2)

(a) $S = 5 \text{ m/s}$
 $t = ?$

$$d = S \times t$$

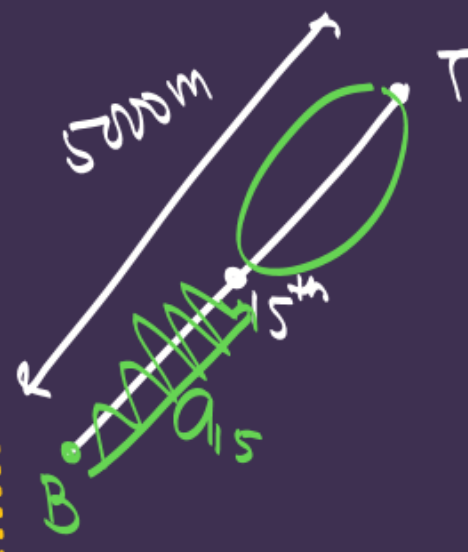
$$t = \frac{d}{S} = \frac{?}{5 \text{ m/s}}$$



dist of 15th pole from top

$$d \Rightarrow 5000 \text{ m} - a_{15}$$

$$d = \bigcirc$$



$a_n \rightarrow$ dist of last pole from bar

$$a_n = 5000 - 300$$

$$a + (n-1)d = 4700$$

$$200 + (n-1)(150) = 4700$$

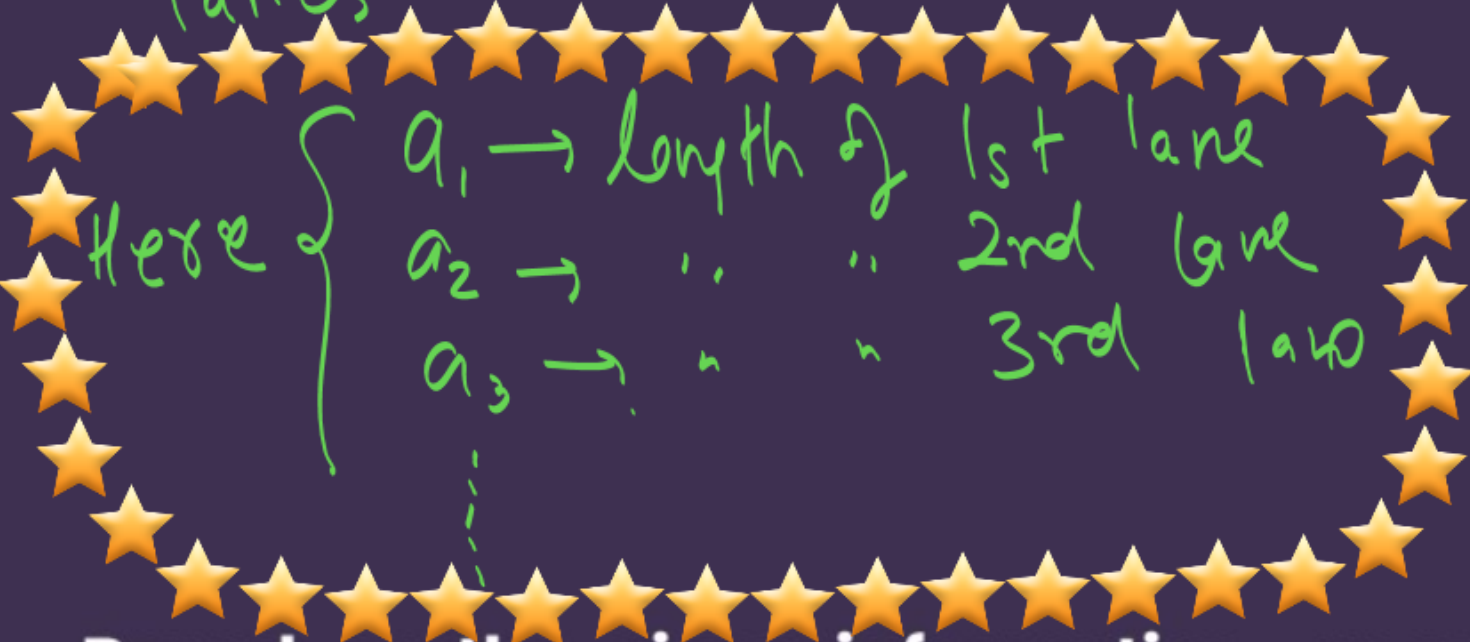


$$n = 31 \text{ poles}$$

#LP: In order to organise, Annual sports Day, a school prepared an eight-lane running track with an integrated football field inside the track area as shown below:

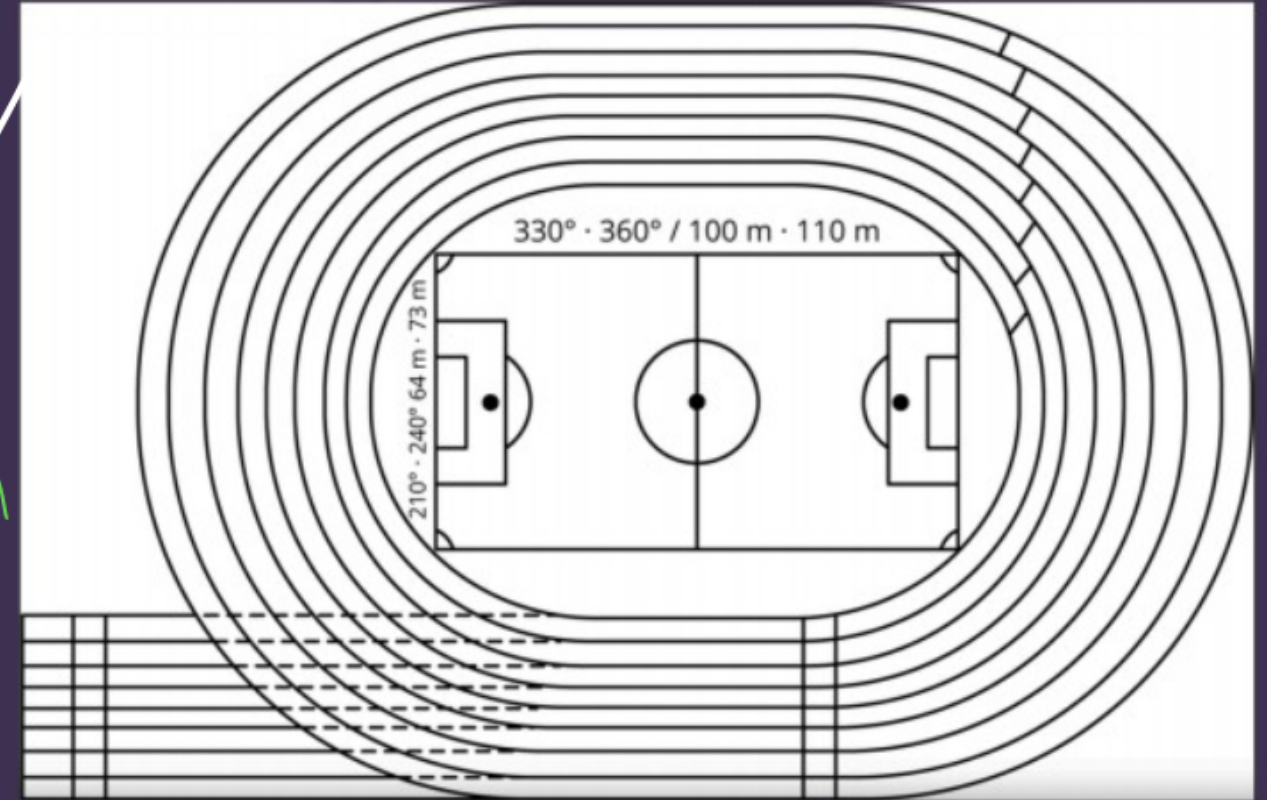
The length of the innermost lane of the track is 400 m and each subsequent lane is 7.6 m longer than the preceding lane. [CBSE 2026]

length of lanes $\rightarrow 400\text{m}, 407.6\text{m}, 415.2\text{m} \dots$ - - - AP



Here $\left\{ \begin{array}{l} a_1 \rightarrow \text{length of 1st lane} \\ a_2 \rightarrow \text{2nd lane} \\ a_3 \rightarrow \text{3rd lane} \end{array} \right.$

$$a = 400\text{m}$$
$$d = 7.6\text{m}$$



Based on the given information, answer the following questions, using the concept of arithmetic progression.

i. What is the length of the 6th lane?

$$a_6 = a + 5d$$

ii. How long is the 8th lane than that of 4th lane?

$$a_8 - a_4$$

length of 8th lane length of 4th lane.

iii. a. While practicing for a race, a student took one round each in first six lanes. Find the total distance covered by the student.

OR

b. A student took one round each in lane 4 to lane 8. Find the total distance covered by the student.

(a) total dist $\rightarrow S_6$ ✓

(b) $S_{4 \rightarrow 8} \Rightarrow S_8 - S_3$

or, $a_4 + a_5 + a_6 + a_7 + a_8$

[Aim: 100/100 in maths]



$$3^{-\pi^2} 2^{\frac{9}{2}} 2^T$$

TRIGONOMETRY

Lecture #1

TRIGONOMETRY

three

sides

measurement

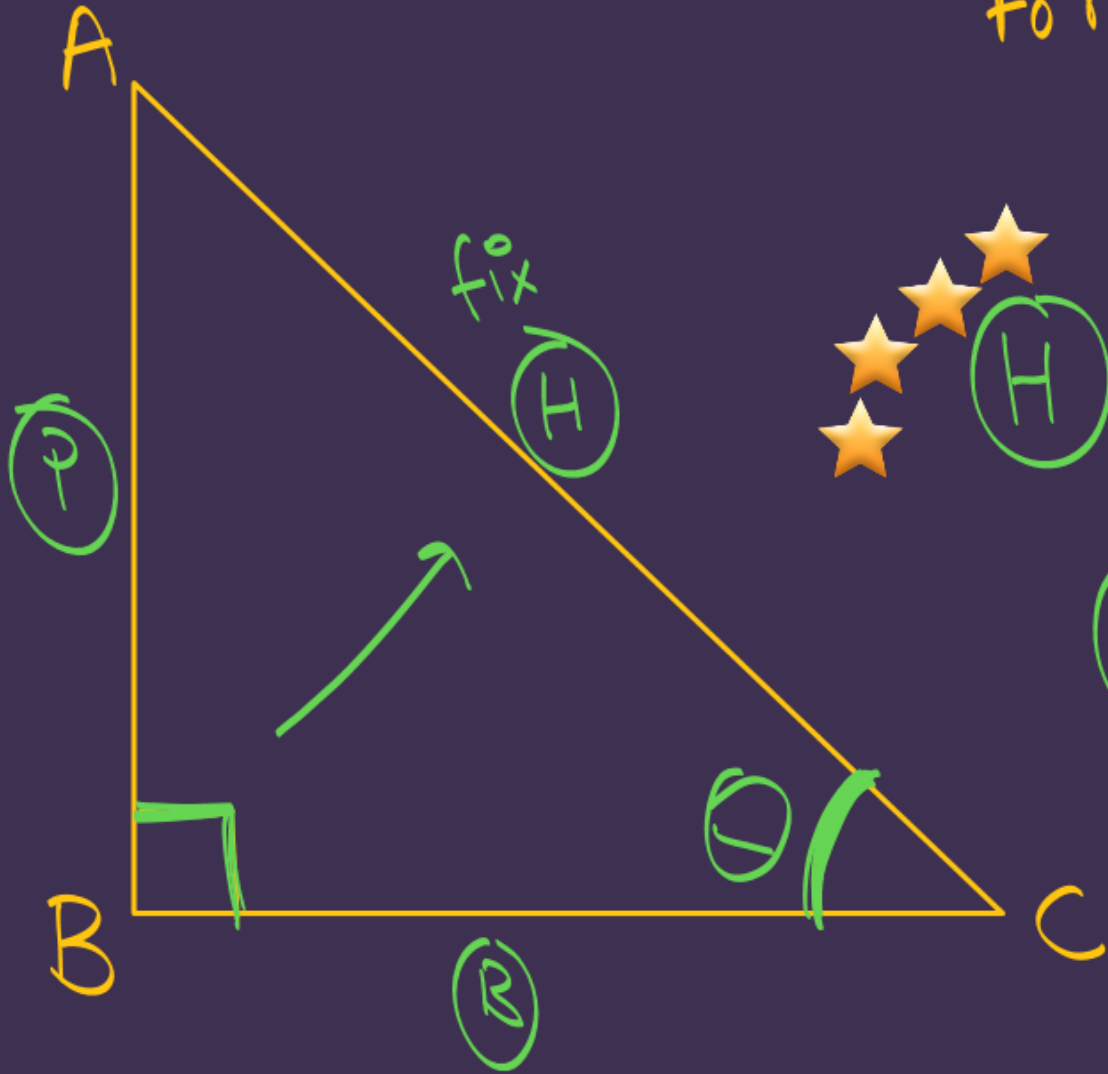
Right angled Δ

for right angled $\Delta \rightarrow$

Pythagoras theorem is valid

$$H^2 = P^2 + B^2$$

$\theta \rightarrow$ theta



$\rightarrow$ right angle के सामने वाली side

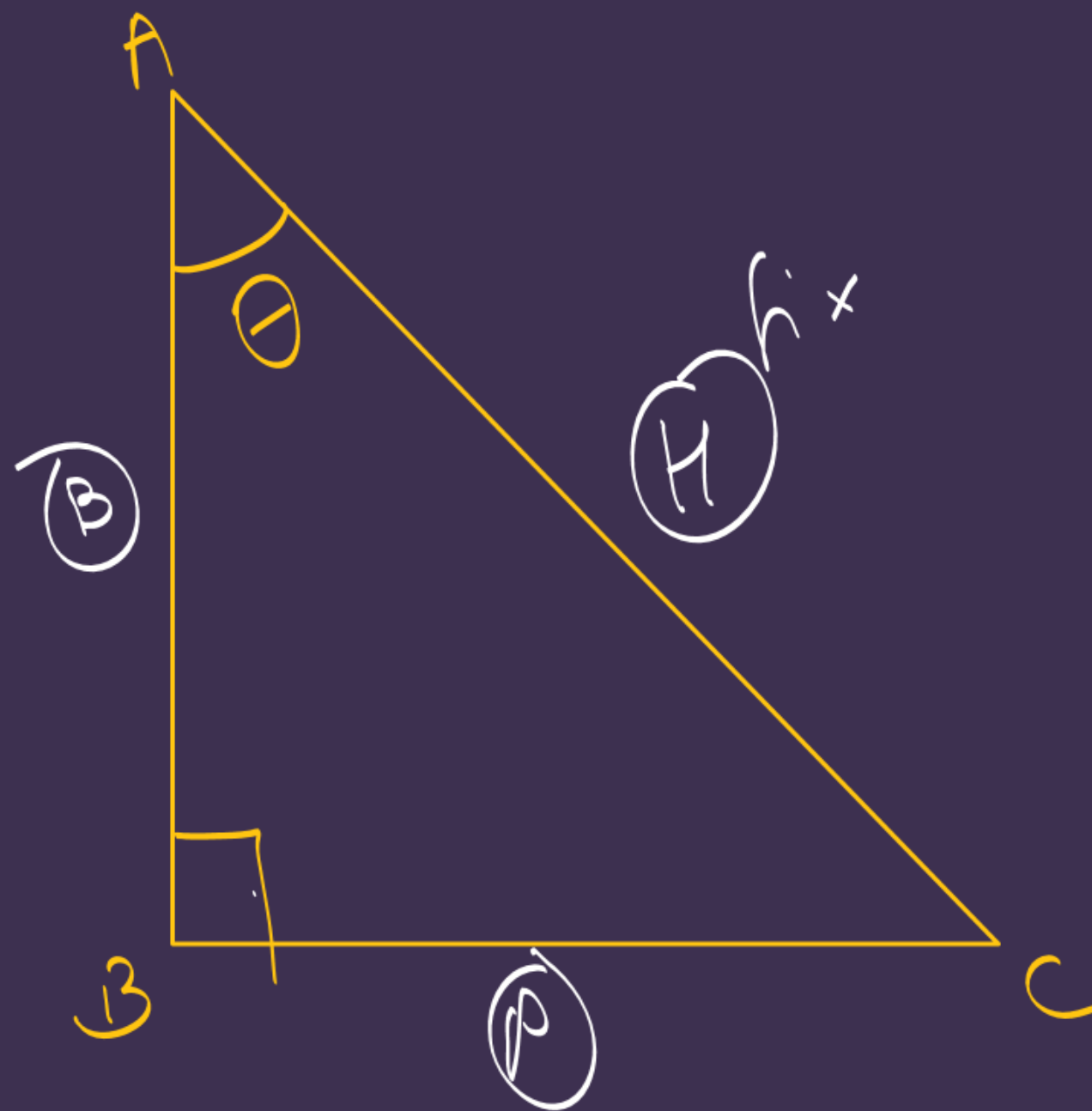
$\textcircled{P}$

$\textcircled{B}$

} depends on angle from which we are seeing

जिस angle की देख दें है $\rightarrow$ उसके सामने वाली side $\textcircled{P}$

जो क्या हो $\rightarrow \textcircled{B}$



$\textcircled{H} \rightarrow AC$

$\textcircled{P} \rightarrow BC$

$\textcircled{B} \rightarrow AB$

Trigonometric Ratios

$$\sin \theta = \frac{p}{H} \longleftrightarrow \operatorname{cosec} \theta = \frac{H}{p}$$

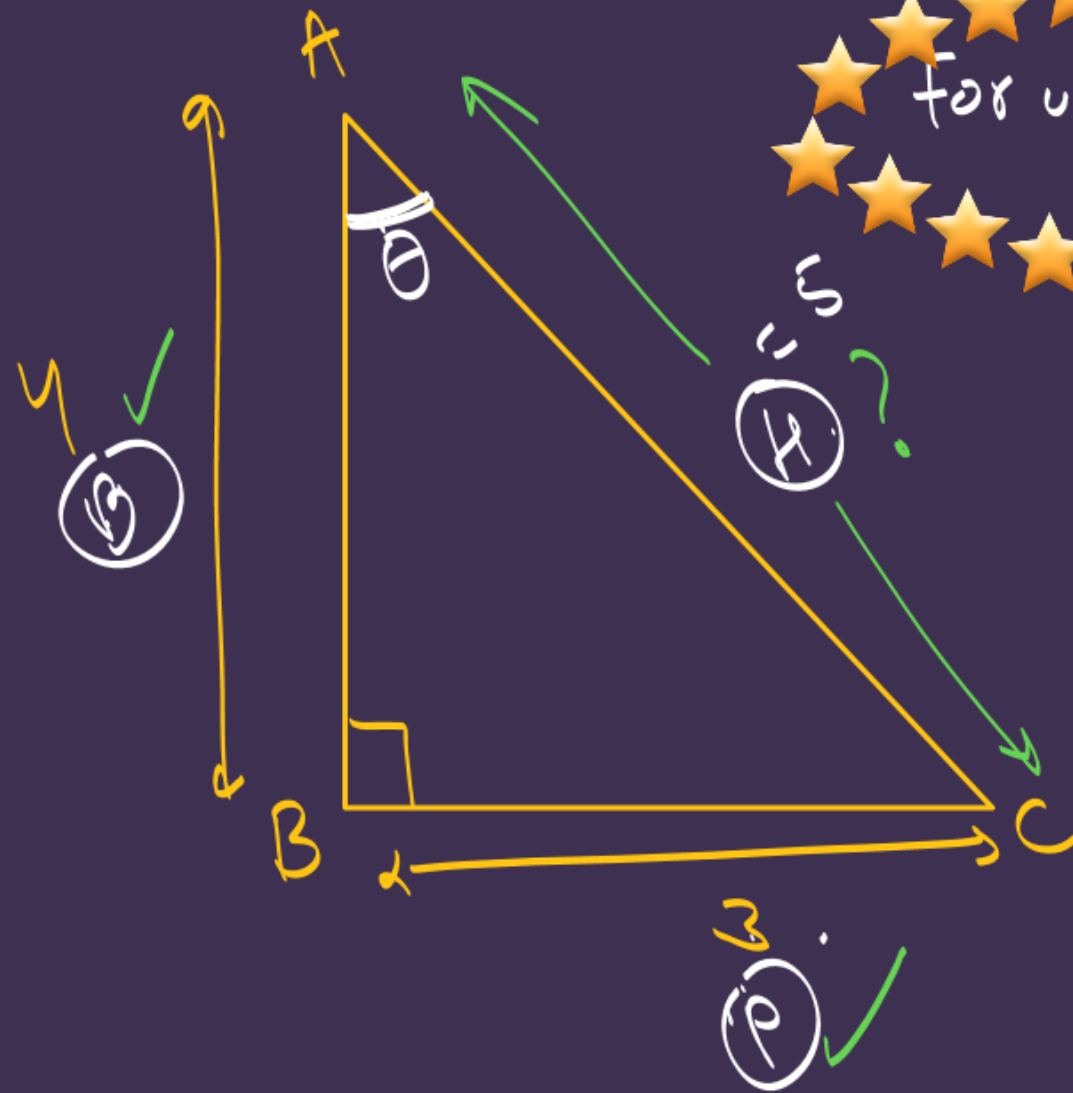
$$\cos \theta = \frac{B}{H} \longleftrightarrow \sec \theta = \frac{H}{B}$$

$$\tan \theta = \frac{p}{B} \longleftrightarrow \cot \theta = \frac{B}{p}$$

shoot trick

$\sin \theta$	$\cos \theta$	$\tan \theta$
P_{apo}	B_{idi}	P_{iyoge}
H_a	H_a	B_{eta}
$\operatorname{cosec} \theta$	$\sec \theta$	$\cot \theta$

#LP: In a triangle ABC, right angled at B, if AB = 4, BC = 3, find all the six trigonometric ratios of angle A.



For unknown side
 $\hookrightarrow$ Pytha

$$(H)^2 = (P)^2 + (B)^2$$

$$(H)^2 = (3)^2 + (4)^2$$

$$H^2 = 9 + 16$$

$$H^2 = 25$$

$$H = \sqrt{25}$$

$$H = \pm 5$$

length cannot be -ve

$$H = +5$$

To find

$$\sin A = \frac{\text{opp}}{\text{hyp}} = \frac{3}{5}$$

$$\cos A = \frac{\text{adj}}{\text{hyp}} = \frac{4}{5}$$

$$\tan A = \frac{\text{opp}}{\text{adj}} = \frac{3}{4}$$

$$\operatorname{cosec} A = \frac{\text{hyp}}{\text{opp}} = \frac{5}{3}$$

$$\sec A = \frac{\text{hyp}}{\text{adj}} = \frac{5}{4}$$

$$\cot A = \frac{\text{adj}}{\text{opp}} = \frac{4}{3}$$

THANK YOU COODIESS !!

